QUAID-I-AZAM UNIVERSITY ISLAMABAD

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DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

SOFTWARE PROJECT TECHNICAL DOCUMENTATION

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Abstract

Detection of clickbait through Machine Learning

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

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Chapter 2

Software Requirement Specification

UC-7 Train Machine Learning Model

USE CASE ID	UC-7
USE CASE NAME	Train Machine Learning Model
Actors	User
Description	It describes how the system trains the model the use of existing facts.
Preconditions	Dataset must be available.
Postconditions	Model is skilled and geared up for use.
Normal Flow	1. Admin selects training option. 2. System approaches dataset. 3. Model is educated. 4. Training finishing touch is showed.
Alternative Flow	**2a.** Insufficient data → training aborted.
Exception	Model training failure.
Special Requirements	Training consequences need to be saved.
Frequency of Use	Low
Technology and Data variation List	Different dataset sizes and features.

Table 7 Use Case Description UC-7 Train Machine Learning Model

UC-8 Detect Clickbait

USE CASE ID	UC-8
USE CASE NAME	Detect Clickbait
Actors	User
Description	It explains how the machine classifies textual content as clickbait or non-clickbait.
Preconditions	Text must be provided.
Postconditions	Prediction result is generated.
Normal Flow	1. User submits headline or information textual content for analysis. 2. The system completes the detection process. 3. The system shows the prediction result to the user.
Alternative Flow	**1a.** Empty input → prompt user.
Exception	Prediction processing error.
Special Requirements	Prediction response time must be reasonable.
Frequency of Use	High.

Technology and Data variation List	Different writing styles and vocabulary.

Table 8 Use Case Description UC-8 Detect Clickbait

UC-9 View Prediction Result

USE CASE ID	UC-9
USE CASE NAME	View Prediction Result
Actors	User
Description	This use case suggests how the device shows effects after textual content submission.
Preconditions	1. User must be logged in. 2. A clickbait detection process must be completed.
Postconditions	Prediction Result is display to the user. The person can understand whether or not the content is classified as clickbait or non-clickbait.
Normal Flow	1. User submits headline or information textual content for analysis. 2. The system completes the detection process. 3. The system shows the prediction result to the user.
Alternative Flow	**2a.** If prediction result is not available, the system informs the user to retry the detection process.
Exception	1. System error during result retrieval. 2. Prediction output failure.
Special Requirements	1. Prediction result must be displayed in a clear and understandable format. 2. Result need to be shown within a suitable response time.
Frequency of Use	High.
Technology and Data variation List	1. Result show may additionally range based on textual content length and category self belief. 2. Different output codecs may be used for result presentation.

Table 9 Use Case Description UC-9 View Prediction Result

UC-10 Explain Prediction

USE CASE ID	UC-10
USE CASE NAME	Explain Prediction
Actors	User
Description	This use case explains why the gadget categorized the textual content as clickbait or no longer.
Preconditions	Prediction must be available.
Postconditions	Explanation is displayed.

Normal Flow	1. User requests rationalization. 2. System shows influential elements.
Alternative Flow	**N/A**
Exception	Explanation generation failure.
Special Requirements	Explanation must be understandable to users.
Frequency of Use	Medium.
Technology and Data variation List	Variation in rationalization element degree.

Table 10 Use Case Description UC-10 Explain Prediction

UC-11 View Model Accuracy and Performance

USE CASE ID	UC-11
USE CASE NAME	Explain Prediction
Actors	User
Description	This use case explains why the machine classified the textual content as clickbait or no longer.
Preconditions	Prediction must be available.
Postconditions	Explanation is displayed.
Normal Flow	1. User requests clarification. 2. System presentations influential factors.
Alternative Flow	**N/A**
Exception	Explanation generation failure.
Special Requirements	Explanation need to be understandable to customers.
Frequency of Use	Medium.
Technology and Data variation List	Variation in explanation detail level.

Table 11 Use Case Description UC-11 View Model Accuracy and Performance

UC-12 Save Detection History

USE CASE ID	UC-12
USE CASE NAME	Save Detection History
Actors	User
Description	This use case saves prediction outcomes for destiny reference.
Preconditions	Prediction must exist.
Postconditions	Detection record is stored.
Normal Flow	1. System saves result automatically.

Alternative Flow	**N/A**
Exception	Storage failure.
Special Requirements	Data integrity should be maintained.
Frequency of Use	High.
Technology and Data variation List	Storage size variation.

Table 12 Use Case Description UC-12 Save Detection History

UC-13 View Detection History

USE CASE ID	UC-13
USE CASE NAME	View Detection History
Actors	User
Description	This use case lets in customers to look previous clickbait detection outcomes.
Preconditions	User must be logged in.
Postconditions	Detection history is displayed.
Normal Flow	1. User opens history page. 2. System retrieves records. 3. History is shown.
Alternative Flow	**2a.** No records → system shows empty state.
Exception	Retrieval error.
Special Requirements	History need to be looked after by way of date.
Frequency of Use	Medium.
Technology and Data variation List	Variable record count.

Table 13 Use Case Description UC-13 View Detection History

System Sequence Diagram

SYSTEM Sequence Diagrams Illustrate the interactions between outside actors and the SYSTEM displaying the waft of events and information over time.

SSD UC-1 User Registration

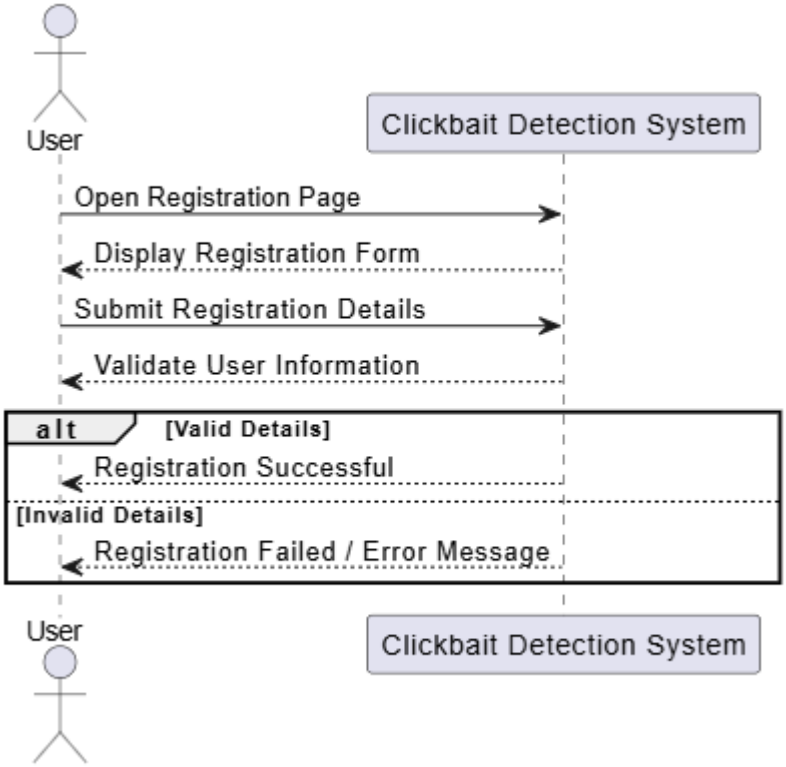


Figure 3 SSD UC-1 User Registration

SSD UC-2 User Login

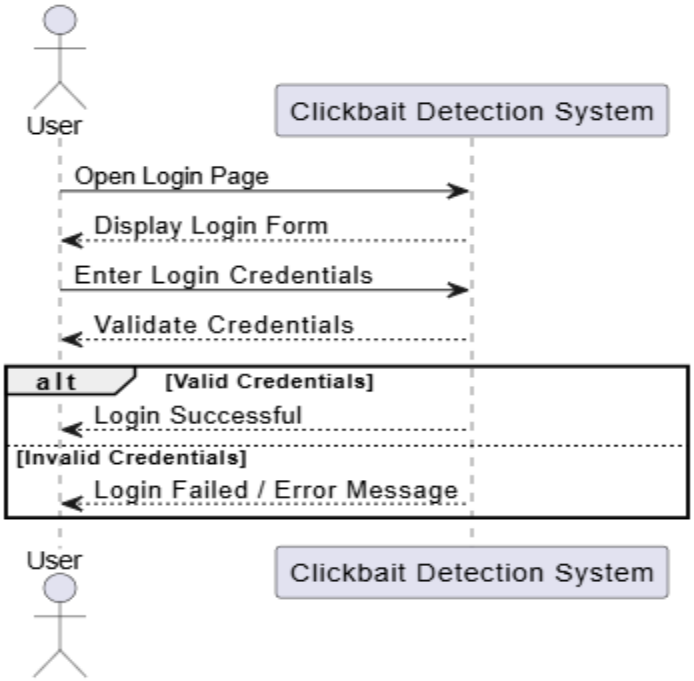


Figure 4 SSD UC-2 User Login

SSD UC-3 Reset Password

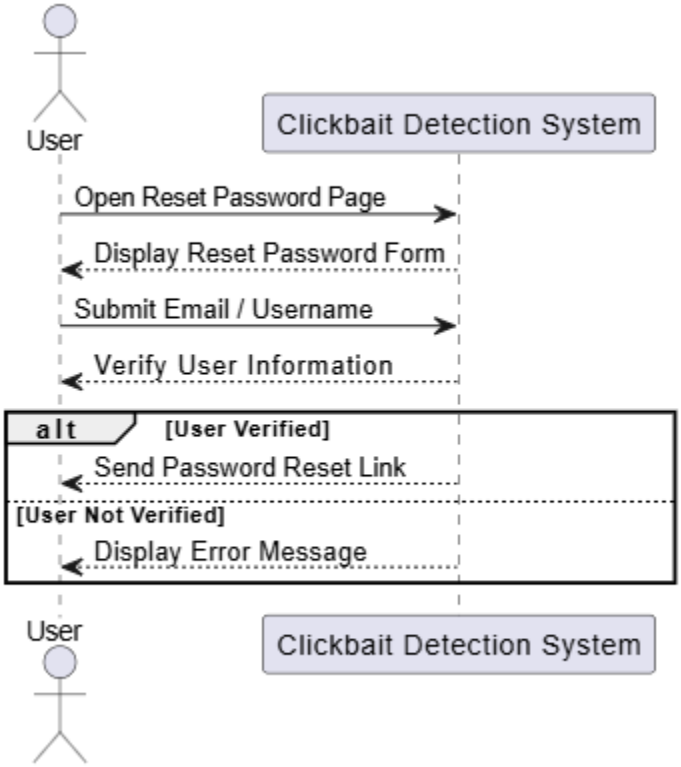


Figure 5 SSD UC-3 Reset Password

SSD UC-4 User Logout

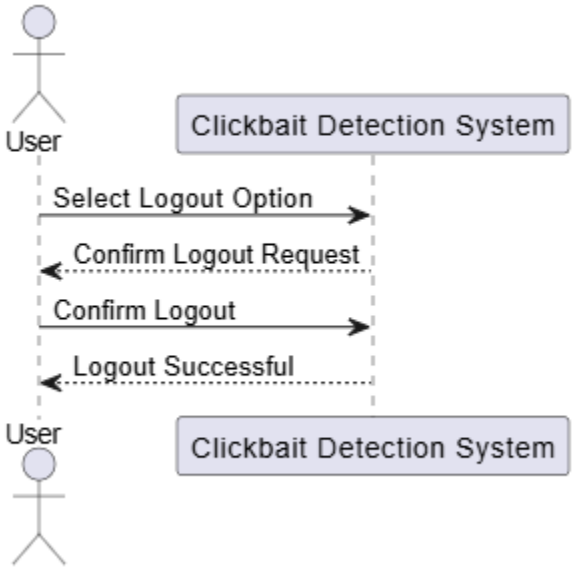


Figure 6 SSD UC-4 User Logout

SSD UC-5 Input Headlines / Text

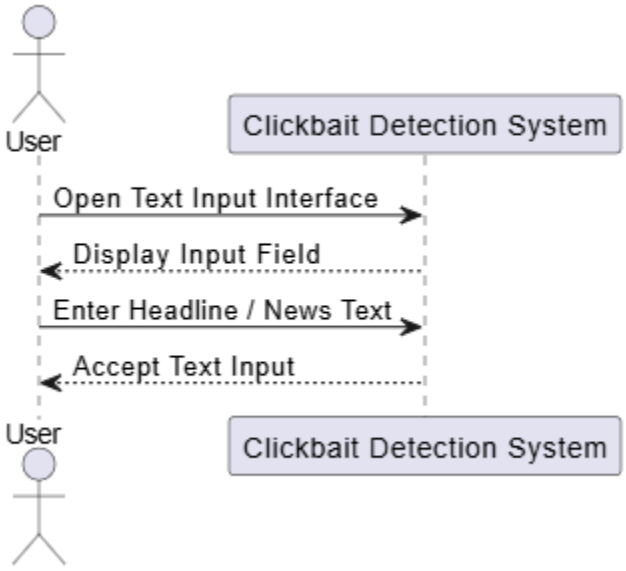


Figure 7 SSD UC-5 Input Headlines / Text

SSD UC-6 Upload Dataset

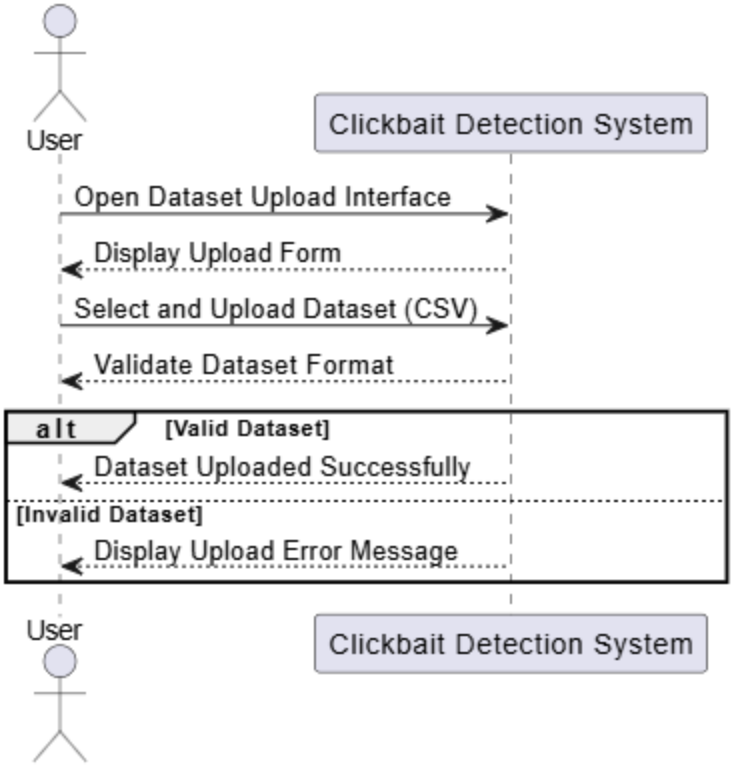


Figure 8 SSD UC-6 Upload Dataset